Evaluation runs: 2pc-ab (two-pc-abstain-kragl)

generated 2026-11-08 from brain-data e1e40f443 2026-11-08 13:42:54 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{ref}, G_{cand} = the graphs of the reference and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G_{ref}^{expl}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕓 = the time tier was reached; oom = the memory limit was reached.

stateless

stateless: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
1	7	3	conforms	0.0	4	1 h, 250 GB	3	14	brain03, 2026-11-03
2	36	13	conforms	0.0	4	1 h, 250 GB	11	171	brain03, 2026-11-03
3	238	67	conforms	0.1	4	1 h, 250 GB	27	883	brain03, 2026-11-03
4	1962	377	conforms	9.9	6	1 h, 250 GB	172	145711	brain03, 2026-11-03
5	19744	2291	conforms	406.5	15	1 h, 250 GB	690	5862635	brain03, 2026-11-03
6	236938	15205	conforms	1022.7	30	1 h, 250 GB	746	12379136	brain03, 2026-11-03
7	3314280		🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-03

stateless: pair **off-by-one**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	sweeps	G_{ref}^{expl}	run
1	7	3	2	violation	0.0	4	1 h, 250 GB	3	20	brain03, 2026-11-03
2	36	13	6	violation	0.1	4	1 h, 250 GB	12	283	brain03, 2026-11-03
3	238	67	18	violation	0.5	4	1 h, 250 GB	56	8825	brain03, 2026-11-03
4	1962	377	72	violation	9.7	4	1 h, 250 GB	148	159953	brain03, 2026-11-03
5	19744	2291	360	violation	751.6	19	1 h, 250 GB	1065	13419655	brain03, 2026-11-03
6	236938			🕓 (1 h)			1 h, 250 GB			brain03, 2026-11-03

Not run (stateless). planned but never scheduled (grid points beyond the ladder): 8 off-by-one bag, 8 correct bag, 8 off-by-one bag.t16.p.r3, 8 correct bag.t16.p.r3, 9 off-by-one bag, 9 correct bag, 9 off-by-one bag.t16.p.r3, 9 correct bag.t16.p.r3, 10 off-by-one bag, 10 correct bag.

skipped: a smaller size had reached its limit: 8 off-by-one bag, 8 correct bag, 9 off-by-one bag, 9 correct bag, 10 off-by-one bag, 10 correct bag, 11 correct bag, 11 off-by-one bag.

left out by the table rules (invalid: foreign load): 7 off-by-one bag.

stateful

stateful: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
1	7	3	conforms	0.0	2	48 h, 250 GB	brain03, 2026-11-06
2	36	13	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-03
3	238	67	conforms	0.0	2	1 h, 250 GB	brain02, 2026-11-03
4	1962	377	conforms	0.2	2	1 h, 250 GB	brain02, 2026-11-03

size	G_{spec}	G_{impl}	verdict	T	M	tier	run
5	19744	2291	conforms	1.5	4	1 h, 250 GB	brain02, 2026-11-03
6	236938	15205	conforms	8.0	44	1 h, 250 GB	brain02, 2026-11-03
7	3314280	111539	conforms	63.3	238	1 h, 250 GB	brain02, 2026-11-03
8	52994722	909809	conforms	1111.8	873	48 h, 250 GB	brain04, 2026-11-05
9	953561800	8247331	conforms	12426.0	3020	48 h, 250 GB	brain04, 2026-11-05
10			(48 h)			48 h, 250 GB	brain04, 2026-11-03

stateful: pair **off-by-one**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	run
1	7	3	2	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
2	36	13	6	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
3	238	67	18	violation	0.0	2	1 h, 250 GB	brain02, 2026-11-03
4	1962	377	72	violation	0.2	2	1 h, 250 GB	brain02, 2026-11-03
5	19744	2291	360	violation	1.2	8	1 h, 250 GB	brain02, 2026-11-03
6	236938	15205	2160	violation	7.1	42	1 h, 250 GB	brain02, 2026-11-03
7	3314280	111539	15120	violation	55.7	242	1 h, 250 GB	brain02, 2026-11-03
8	52994722	909809	120960	violation	457.1	838	48 h, 250 GB	brain04, 2026-11-05
9	953561800	8247331	1088640	violation	11247.0	3025	48 h, 250 GB	brain04, 2026-11-05
10				(48 h)			48 h, 250 GB	brain04, 2026-11-03

Not run (stateful). left out by the table rules (ok): 1 correct bag, 1 correct bag.t16.c48h.r7, 8 off-by-one bag, 8 correct bag, 8 correct bag.t16.r41.

the tool ended with an error (exit-143): 8 correct bag.t16.r42.

planned but never scheduled (grid points beyond the ladder): 8 off-by-one bag.t16.p.r3, 8 correct bag.t16.p.r3, 8 correct bag.t16.r31, 8 correct bag.t16.r32, 8 correct bag.t16.r33, 8 correct bag.t16.r34, 8 correct bag.t16.r35, 8 correct bag.t16.r43, 8 correct bag.t16.r44, 8 correct bag.t16.r45, 11 correct bag, 11 off-by-one bag, 12 off-by-one bag, 13 correct bag, 13 off-by-one bag, 14 off-by-one bag, 14 correct bag, 15 off-by-one bag, 15 correct bag, 16 correct bag, 16 off-by-one bag.

left out by the table rules (timeout): 9 correct bag, 9 off-by-one bag.

skipped: a smaller size had reached its limit: 10 correct bag, 10 off-by-one bag, 11 correct bag, 11 off-by-one bag.

left out by the table rules (invalid: foreign load): 11 off-by-one bag, 12 correct bag.

offline

offline: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	7	3	conforms	0.1	8	48 h, 250 GB	0.0	0.1	0.0	merge	brain03, 2026-11-06
2	36	13	conforms	0.0	8	4 h, 250 GB	0.0	0.0	0.0	merge	brain05, 2026-11-03
3	238	67	conforms	0.1	8	4 h, 250 GB	0.1	0.0	0.0	merge	brain05, 2026-11-03
4	1962	377	conforms	0.3	8	4 h, 250 GB	0.3	0.1	0.0	merge	brain05, 2026-11-03
5	19744	2291	conforms	1.4	8	4 h, 250 GB	1.2	0.3	0.2	merge	brain05, 2026-11-03
6	236938	15205	conforms	8.0	46	4 h, 250 GB	7.1	1.3	0.9	merge	brain05, 2026-11-03
7	3314280	111539	conforms	83.9	226	4 h, 250 GB	78.2	4.3	5.7	merge	brain05, 2026-11-03
8	52994722	909809	conforms	1414.7	1133	4 h, 250 GB	1381.7	29.0	33.1	merge	brain05, 2026-11-03
8	52994722	909809	conforms	1257.2	1051	48 h, 250 GB	1231.1	17.3	26.2	merge	brain04, 2026-11-05
9	953561800	8247331	conforms	11364.9	5887	48 h, 250 GB	11234.0	109.3	130.9	merge	brain03, 2026-11-03
9	953561800	8247331	conforms	12702.9	5811	48 h, 250 GB	12560.0	106.8	142.9	merge	brain04, 2026-11-06

offline: pair **off-by-one**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	7	3	2	violation	0.2	8	4 h, 250 GB	0.2	0.0	0.0	merge	brain05, 2026-11-03
2	36	13	6	violation	0.1	8	4 h, 250 GB	0.0	0.1	0.0	merge	brain05, 2026-11-03
3	238	67	9	violation	0.1	8	4 h, 250 GB	0.1	0.1	0.0	merge	brain05, 2026-11-03
4	1962	377	12	violation	0.3	8	4 h, 250 GB	0.3	0.1	0.0	merge	brain05, 2026-11-03
5	19744	2291	15	violation	1.4	10	4 h, 250 GB	1.2	0.4	0.2	merge	brain05, 2026-11-03
6	236938	15205	18	violation	8.0	44	4 h, 250 GB	7.1	1.0	0.9	merge	brain05, 2026-11-03
7	3314280	111539	21	violation	83.8	232	4 h, 250 GB	78.2	4.5	5.5	merge	brain05, 2026-11-03

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
8	52994722	909809	24	violation	1415.4	1135	4 h, 250 GB	1381.7	26.6	33.8	merge	brain05, 2026-11-03
8	52994722	909809	24	violation	1256.2	1044	48 h, 250 GB	1231.1	12.6	25.1	merge	brain04, 2026-11-05
9	953561800	8247331	27	violation	11366.2	5884	48 h, 250 GB	11234.0	121.4	132.2	merge	brain03, 2026-11-03
9	953561800	8247331	27	violation	12702.9	5808	48 h, 250 GB	12560.0	101.1	142.9	merge	brain04, 2026-11-06

Not run (offline). planned but never scheduled (grid points beyond the ladder): 8 off-by-one bag.p.r3, 8 correct bag.p.r3, 8 correct bag.r31, 8 correct bag.r32, 8 correct bag.r33, 8 correct bag.r34, 8 correct bag.r35, 8 correct bag.r42, 8 correct bag.r43, 8 correct bag.r44, 8 correct bag.r45, 11 correct bag, 11 off-by-one bag.

left out by the table rules (ok): 1 correct bag, 1 correct bag.m.c48h.r7, 8 correct bag.r41, 9 off-by-one bag, 9 correct bag.

refused (an input missing at run time): 10 correct bag, 10 off-by-one bag.

skipped: a smaller size had reached its limit: 11 correct bag, 11 off-by-one bag.

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	7	3	conforms	2.7	153	48 h, 250 GB	0.0	2.7	0.0	merge	brain03, 2026-11-06
2	36	13	conforms	6.2	154	4 h, 250 GB	0.0	6.2	0.0	merge	brain05, 2026-11-03
3	238	67	conforms	6.7	154	4 h, 250 GB	0.1	6.7	0.0	merge	brain05, 2026-11-03
4	1962	377	conforms	4.3	170	10 min, 16 GB	0.2	4.2	0.1	in-memory	brain01, 2026-11-02
5	19744	2291	conforms	7.9	492	10 min, 16 GB	1.0	7.7	0.1	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **off-by-one**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	7	3	2	violation	6.0	160	4 h, 250 GB	0.2	6.0	0.0	merge	brain05, 2026-11-03
2	36	13	6	violation	6.4	160	4 h, 250 GB	0.0	6.4	0.0	merge	brain05, 2026-11-03
3	238	67	9	violation	6.9	167	4 h, 250 GB	0.1	6.9	0.0	merge	brain05, 2026-11-03
4	1962	377	12	violation	4.2	174	10 min, 16 GB	0.2	4.2	0.0	in-memory	brain01, 2026-11-02
5	19744	2291	15	violation	7.9	473	10 min, 16 GB	1.0	7.7	0.2	in-memory	brain01, 2026-11-02

Not run (cross offline, Java implementation against the Rust specification). planned but never scheduled (grid points beyond the ladder): 6 off-by-one bag, 6 correct bag, 8 off-by-one bag, 8 correct bag, 9 off-by-one bag, 9 correct bag, 10 correct bag.

left out by the table rules (ok): 1 correct bag, 1 correct bag.tfspec-jmcimpl.m.c48h.r7.

skipped: a smaller size had reached its limit: 10 off-by-one bag.

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	7	3	conforms	6.2	170	4 h, 250 GB	6.2	0.0	0.0	merge	brain05, 2026-11-03
2	36	13	conforms	6.7	154	4 h, 250 GB	6.7	0.0	0.0	merge	brain05, 2026-11-03
3	238	67	conforms	8.4	159	4 h, 250 GB	8.4	0.0	0.0	merge	brain05, 2026-11-03
4	1962	377	conforms	19.6	1932	4 h, 250 GB	19.6	0.1	0.0	merge	brain05, 2026-11-03
5	19744	2291	conforms	118.0	2970	4 h, 250 GB	117.8	0.3	0.1	merge	brain05, 2026-11-03

cross offline, Rust implementation against the Java specification: pair **off-by-one**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	7	3	2	violation	6.2	170	4 h, 250 GB	6.2	0.0	0.0	merge	brain05, 2026-11-03
2	36	13	6	violation	6.7	154	4 h, 250 GB	6.7	0.1	0.0	merge	brain05, 2026-11-03
3	238	67	9	violation	8.4	159	4 h, 250 GB	8.4	0.1	0.0	merge	brain05, 2026-11-03
4	1962	377	12	violation	19.6	1932	4 h, 250 GB	19.6	0.1	0.0	merge	brain05, 2026-11-03
5	19744	2291	15	violation	75.6	1656	10 min, 16 GB	75.4	0.2	0.1	in-memory	brain04, 2026-11-01

Not run (cross offline, Rust implementation against the Java specification). planned but never scheduled (grid points beyond the ladder): 6 correct bag, 8 off-by-one bag, 8 correct bag, 9 off-by-one bag, 9 correct bag.

refused (an input missing at run time): 10 correct bag, 10 off-by-one bag.

cross online, Java implementation (JMC) against the Rust specification export

cross online, Java implementation (JMC) against the Rust specification export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	7	3	conforms	2.8	148	48 h, 250 GB		brain03, 2026-11-06
2	36	13	conforms	0.8	175	1 h, 250 GB		brain05, 2026-11-02
3	238	67	conforms	1.0	166	1 h, 250 GB		brain05, 2026-11-02
4	1962	377	conforms	2.2	287	1 h, 250 GB		brain05, 2026-11-02
5	19744	2291	conforms	10.2	864	1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **off-by-one**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1	7	3	2	violation	0.7	151	1 h, 250 GB	74	brain05, 2026-11-02
2	36	13	6	violation	0.7	146	1 h, 250 GB	42	brain05, 2026-11-02
3	238	67	18	violation	1.1	163	1 h, 250 GB	58	brain05, 2026-11-02
4	1962	377	72	violation	2.1	275	1 h, 250 GB	50	brain05, 2026-11-02
5	19744	2291	360	violation	10.8	947	1 h, 250 GB	54	brain05, 2026-11-02

Not run (cross online, Java implementation (JMC) against the Rust specification export). planned but never scheduled (grid points beyond the ladder): 1 off-by-one bag, 1 correct bag, 2 off-by-one bag, 2 correct bag, 3 off-by-one bag, 3 correct bag, 4 correct bag, 4 off-by-one bag, 5 correct bag, 5 off-by-one bag, 8 off-by-one bag, 8 correct bag, 9 off-by-one bag, 9 correct bag, 10 off-by-one bag, 10 correct bag.

left out by the table rules (ok): 1 correct bag, 1 correct bag.w16.c48h.r7.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **correct**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	7	3	conforms	3.1	156	1 h, 250 GB		brain04, 2026-11-01
2	36	13	conforms	3.4	156	1 h, 250 GB		brain04, 2026-11-01
3	238	67	conforms	4.0	170	1 h, 250 GB		brain04, 2026-11-01
4	1962	377	conforms	9.3	753	1 h, 250 GB		brain04, 2026-11-01
5	19744	2291	conforms	73.1	1733	1 h, 250 GB		brain04, 2026-11-01

cross online, Rust implementation (TraceForge) against the Java specification export: pair off-by-one, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1	7	3	2	violation	3.1	156	1 h, 250 GB	1	brain04, 2026-11-01
2	36	13	6	violation	3.4	156	1 h, 250 GB	2	brain04, 2026-11-01
3	238	67	18	violation	4.0	170	1 h, 250 GB	3	brain04, 2026-11-01
4	1962	377	72	violation	9.3	753	1 h, 250 GB	3	brain04, 2026-11-01
5	19744	2291	360	violation	73.1	1733	1 h, 250 GB	27	brain04, 2026-11-01

Not run (cross online, Rust implementation (TraceForge) against the Java specification export). planned but never scheduled (grid points beyond the ladder): 8 off-by-one bag, 8 correct bag, 9 off-by-one bag, 9 correct bag, 10 off-by-one bag, 10 correct bag.

left out by the table rules (ok): 1 off-by-one bag, 1 correct bag, 2 off-by-one bag, 2 correct bag, 3 off-by-one bag, 3 correct bag, 4 correct bag, 4 off-by-one bag, 5 correct bag, 5 off-by-one bag.